Introduction to Artificial Intelligence and Machine Learning (CSN11601)

ASSIGNMENT-2

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Q1. Create a list and perform the following methods

a) insert()
b) remove()
c) append()
d) len()
e) pop()
f) clear().

CODE :

```
l1=[12,48,51,56,59,43,13,235,124]
print("The list : ")
print(l1)
print("----------INSERT FUNCTION----------")
a=int(input("Enter the value you want to insert : "))
b=int(input("Enter the index at where you want to input the value : "))
l1.insert(b,a)
print("Element Inserted")
print(l1)
print("----------REMOVE FUNCTION----------")
c=int(input("Enter the value you want to remove : "))
l1.remove(c)
print("Element Removed")
print(l1)
print("----------APPEND FUNCTION----------")
d=int(input("Enter the value you want to append : "))
l1.append(d)
print("Element added at last")
print(l1)
print("----------LENGTH FUNCTION----------")
print("Length of the list is : ",len(l1))
print("----------POP FUNCTION----------")
print("Element Poped : ",l1.pop())
print(l1)
print("----------CLEAR FUNCTION----------")
l1.clear()
print("List Cleared : ")
print(l1)
```

OUTPUT :

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignme
nt-2$ python3 p1.py
The list :
[12, 48, 51, 56, 59, 43, 13, 235, 124]
-----INSERT FUNCTION-----
Enter the value you want to insert : 67
Enter the index at where you want to input the value : 4
Element Inserted
[12, 48, 51, 56, 67, 59, 43, 13, 235, 124]
-----REMOVE FUNCTION-----
Enter the value you want to remove : 51
Element Removed
[12, 48, 56, 67, 59, 43, 13, 235, 124]
-----APPEND FUNCTION-----
Enter the value you want to append : 66
Element added at last
[12, 48, 56, 67, 59, 43, 13, 235, 124, 66]
-----LENGTH FUNCTION-----
Length of the list is :  10
-----POP FUNCTION-----
Element Poped :  66
[12, 48, 56, 67, 59, 43, 13, 235, 124]
-----CLEAR FUNCTION-----
List Cleared :
[]
```

Q2. Write a Python function that takes two lists and returns True if they are equal otherwise false.

A2. CODE :

```
def test_list(l1,l2):
    return l1==l2
list1=[]
list2=[]
print("For list 1")
n1=int(input("Enter the length of list : "))
for i in range(n1):
    list1.append(int(input(f"Enter the {i+1} element : ")))
print("List 1 completed")
print("For list 2")
n2=int(input("Enter the length of list : "))
for i in range(n2):
    list2.append(int(input(f"Enter the {i+1} element : ")))
print("list 2 completed")
print("Final output : ",test_list(list1,list2))
```

OUTPUT :

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignme
nt-2$ python3 p2.py
For list 1
Enter the length of first list : 3
Enter the 1 element : 1
Enter the 2 element : 2
Enter the 3 element : 3
List 1 completed
For list 2
Enter the length of list : 3
Enter the 1 element : 1
Enter the 2 element : 2
Enter the 3 element : 3
list 2 completed
True
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignme
nt-2$ python3 p2.py
For list 1
Enter the length of list : 4
Enter the 1 element : 1
Enter the 2 element : 2
Enter the 3 element : 3
Enter the 4 element : 4
List 1 completed
For list 2
Enter the length of list : 3
Enter the 1 element : 5
Enter the 2 element : 6
Enter the 3 element : 7
list 2 completed
Final output :  False
```

Q3. Create a tuple and perform the following methods

a) Add items
b) len()
c) check for item in tuple
d) Access items.

A3. CODE :

```
t1=(10,20,30,40)
l1=list(t1)
l1.append(50)
t1=tuple(l1)
print("Tuple after adding item:",t1)
print("Length of tuple:", len(t1))
print("Is 30 in tuple?", 30 in t1)
print("Is 100 in tuple?", 100 in t1)
print("First item:", t1[0])
print("Last item:", t1[-1])
print("Slice [1:4]:", t1[1:4])
```

OUTPUT :

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignme
nt-2$ python3 p3.py
Tuple after adding item: (10, 20, 30, 40, 50)
Length of tuple: 5
Is 30 in tuple? True
Is 100 in tuple? False
First item: 10
Last item: 50
Slice [1:4]: (20, 30, 40)
```

Q4. Write a Python program to enter name and percentage marks in a dictionary and display information on the screen.

A4. CODE:

```
students = {}
n = int(input("Enter number of students: "))
for i in range(n):
    name = input(f"Enter name of student {i+1}: ")
    percentage = float(input(f"Enter percentage of {name}: "))
    students[name] = percentage
print("\nStudent Information:")
for name, percentage in students.items():
    print(f"Name: {name}, Percentage: {percentage}%")
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignme
nt-2$ python3 p4.py
Enter number of students: 4
Enter name of student 1: Aayansh
Enter percentage of Aayansh: 98.2
Enter name of student 2: OM Upadhyay
Enter percentage of OM Upadhyay: 96.4
Enter name of student 3: Param Hansh
Enter percentage of Param Hansh: 97.4
Enter name of student 4: Ankur
Enter percentage of Ankur: 96.4

Student Information:
Name: Aayansh, Percentage: 98.2%
Name: OM Upadhyay, Percentage: 96.4%
Name: Param Hansh, Percentage: 97.4%
Name: Ankur, Percentage: 96.4%
```

Q5. Write a program print duplicate element present in the list

A5. CODE :

```
list1 = []
n1=int(input("Enter the length of list : "))
for i in range(n1):
    list1.append(int(input(f"Enter the {i+1} element : ")))
duplicates = []
for item in list1:
    if list1.count(item) > 1 and item not in duplicates:
        duplicates.append(item)
print("Duplicate elements:", duplicates)
```

OUTPUT :

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignme
nt-2$ python3 p5.py
Enter the length of list : 5
Enter the 1 element : 10
Enter the 2 element : 20
Enter the 3 element : 10
Enter the 4 element : 40
Enter the 5 element : 20
Duplicate elements: [10, 20]
```